

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)**

Department of Artificial Intelligence and Data Science

DSA0503 – Query Processing For Data Science

CO1 - Assessment Tool 1 - Data Interpretation

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Missing Data Handling

Given a CSV file containing student marks with missing values.

Task

- **Identify missing values.**
- **Replace missing marks with the column average.**
- **Display the cleaned dataset.**

Aim

To identify missing values in a student marks dataset, replace the missing marks with the corresponding column average, and display the cleaned dataset using Python and Pandas.

Procedure

1. Import the **Pandas** library.
2. Read the CSV file containing student marks using `pd.read_csv()`.
3. Identify the missing values using the `isnull()` function.
4. Calculate the average (mean) of each numerical column.
5. Replace the missing values with the corresponding column average using `fillna()`.
6. Display the cleaned dataset after replacing the missing values.

Input Table (student_marks.csv)

Name	Math	Science	English
Alice	85	90	88
Bob	78	NaN	80
Charlie	92	85	NaN
David	NaN	88	75
Eva	80	79	82

Code:

Program 1: Identify Missing Values

```
import pandas as pd
```

```

# Read the CSV file

df = pd.read_csv("student_marks.csv")

# Display the dataset

print("Student Marks Dataset:")

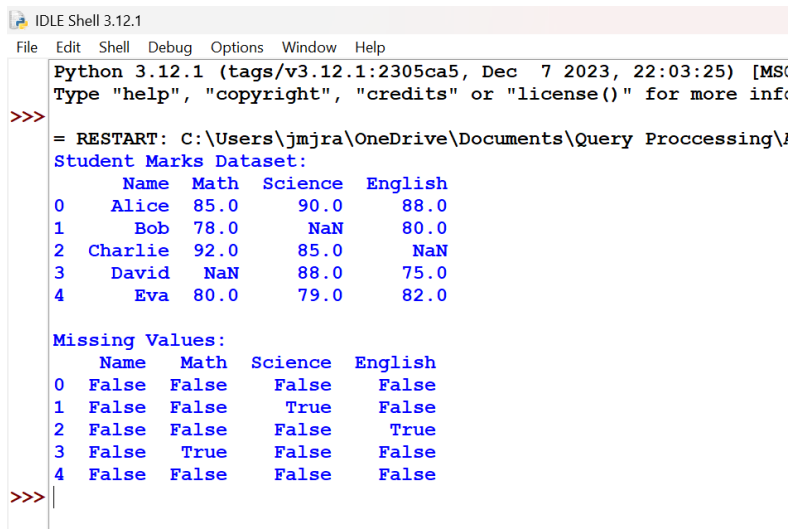
print(df)

# Identify missing values

print("\nMissing Values:")

print(df.isnull())

```



```

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>>>
= RESTART: C:\Users\jnmjra\OneDrive\Documents\Query Processing\l
Student Marks Dataset:
   Name  Math  Science  English
0  Alice  85.0    90.0    88.0
1   Bob   78.0     NaN    80.0
2 Charlie  92.0    85.0     NaN
3  David   NaN    88.0    75.0
4   Eva   80.0    79.0    82.0

Missing Values:
   Name  Math  Science  English
0  False  False   False   False
1  False  False    True   False
2  False  False   False    True
3  False   True   False   False
4  False  False   False   False
>>>

```

Code:

Program 2: Replace Missing Marks with Column Average

```

import pandas as pd

# Read the CSV file

df = pd.read_csv("student_marks.csv")

# Replace missing values with column mean

df.fillna(df.mean(numeric_only=True), inplace=True)

# Display updated dataset

print("Dataset after Replacing Missing Values:")

print(df)

```

```
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>>>
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Dataset after Replacing Missing Values:
      Name  Math  Science  English
0   Alice  85.00    90.0    88.00
1     Bob  78.00    85.5    80.00
2  Charlie  92.00    85.0    81.25
3   David  83.75    88.0    75.00
4     Eva  80.00    79.0    82.00
>>>
```

Code:

Program 3: Display the Cleaned Dataset

```
import pandas as pd

# Read the CSV file

df = pd.read_csv("student_marks.csv")

# Replace missing values with column average

df.fillna(df.mean(numeric_only=True), inplace=True)

# Display cleaned dataset

print("Cleaned Student Marks Dataset:")

print(df)
```

```
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Type "help", "copyright", "credits" or "license()" for
>>>
= RESTART: C:\Users\jnjra\OneDrive\Documents\Query Pro
Cleaned Student Marks Dataset:
      Name  Math  Science  English
0   Alice  85.00    90.0    88.00
1     Bob  78.00    85.5    80.00
2  Charlie  92.00    85.0    81.25
3   David  83.75    88.0    75.00
4     Eva  80.00    79.0    82.00
>>>
```